

Simulation of sound wave transmission in heterogeneous solid media to understand UPV of concrete

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by

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Introduction –

Concrete is one of the most widely used construction materials due to its strength, durability, and versatility. Ensuring the quality and integrity of concrete structures is essential for their safety and long-term performance. In this regard, non-destructive testing (NDT) methods are widely used, as they allow evaluation of material properties without damaging the structure. Among these methods, Ultrasonic Pulse Velocity (UPV) testing is commonly employed to assess the quality, uniformity, and internal condition of concrete.

In a UPV test, an ultrasonic pulse is transmitted through a concrete member, and the time it takes for the wave to travel between two locations is recorded. The velocity is then calculated by dividing the distance by the travel time. Generally, higher velocities indicate dense, high-quality concrete, while lower velocities may suggest the presence of voids, cracks, or insufficient compaction. However, interpreting UPV data can be challenging due to the concrete's heterogeneous nature.

Concrete is composed of multiple phases, including aggregates, mortar, interfacial transition zones (ITZ), and pores. As ultrasonic waves travel through this complex medium, they experience scattering, reflection, and attenuation due to variations in material properties. These interactions can impact the wave's path and travel time, leading to an apparent velocity that may differ from the true material velocity. Therefore, the results of Ultrasonic Pulse Velocity (UPV) testing are influenced not only by the intrinsic properties of the concrete but also by its internal structure and the behaviour of wave propagation within it.

Experimental Ultrasonic Pulse Velocity (UPV) testing provides limited insight into concrete's internal mechanisms, as it captures only the input and output signals. To better understand the effects of heterogeneity, numerical simulations of ultrasonic wave propagation can be used. In this project, a mesoscale model of concrete is developed that explicitly represents aggregates within a mortar matrix. Ultrasonic wave transmission is simulated using transient dynamic analysis to examine wave behaviour in a controlled environment.

The objective of this study is to explore how concrete heterogeneity influences wave propagation and Ultrasonic Pulse Velocity (UPV) measurements. By doing so, we aim to enhance the interpretation of UPV results and emphasise its limitations as a method for assessing concrete quality.

Research Motivation and Objectives

Although Ultrasonic Pulse Velocity (UPV) is a prominent non-destructive characterisation of concrete quality, its interpretation is essentially constrained by the fact that it measures wave travel time rather than an inherent material feature. Ultrasonic waves interact with aggregates, pores, and surfaces in heterogeneous materials such as concrete, leading to scattering, multiple wave paths, and distortion of the first-arrival signal. This indicates that the interior geometry and scale of heterogeneity have the same impact on the observed UPV as the material's elastic characteristics. In addition to calculating UPV from simulations, the goal is to determine the circumstances in which UPV can accurately depict material behaviour as opposed to when it becomes deceptive, as well as to analyse how and why it deviates from theoretical wave velocity.

The primary objective of this research is to go beyond straightforward empirical correlations and create a physics-based understanding of what UPV genuinely signifies in heterogeneous concrete. The main goals are to compare simulated results with experimental UPV measurements and to simulate ultrasonic wave propagation in a mesoscale concrete model that explicitly reflects the function of aggregates and material heterogeneity. The study also seeks to determine the circumstances in which UPV provides accurate or inaccurate information on concrete quality, analyse how UPV differs from theoretical wave velocity, and assess its relationship with elastic parameters such as Young's modulus.

Experimental Results –

An experimental study was conducted in the lab to confirm and supplement the theoretical and numerical understanding of ultrasonic wave propagation in concrete. The desired M30 mix proportions were used to produce 150 mm × 150 mm × 150 mm concrete cubes containing approximately 41% coarse aggregate and 59% mortar by volume.

Table 1 Material and Specimen Details

Grade of concrete	M30
Cube Size	150 x 150 x 150
Type of cement	OPC 43 Grade
Maximum Aggregate size	20 mm
Water – cement ratio	0.43
Cement Content	458 kg/m ³
Fine Aggregate – SSD	602 kg/m ³
Coarse Aggregate - SSD	1117 kg/m ³
Water	197 kg/m ³
Sp. Gravity - Cement	3.15
Sp. Gravity - Fine aggregate	2.54
Sp. Gravity - Coarse aggregate	2.72

$$UPV = \frac{Path\ Length}{Transmission\ Time}$$

After proper curing for 28 days, the specimens were subjected to Ultrasonic Pulse Velocity (UPV) testing to measure the travel time of ultrasonic waves through the concrete. And we got results for the tested blocks: the first block took 43.9 μs, and the second, 39.8 μs.

Table 2 UPV of Concrete Blocks

Block no.	Length	Time	UPV
1.	150 mm	43.9 μs	3416.85 m/s
2.	150 mm	39.8 μs	3768.84 m/s

Simulation Results –

ANSYS was used to perform a numerical simulation to understand the impact of material heterogeneity on ultrasonic wave propagation. Coarse aggregates were explicitly represented within a mortar matrix to create a mesoscale model of concrete. To calculate the corresponding wave travel time and apparent velocity, the simulation mimics the transmission of an ultrasonic wave through concrete.

Model Development –

Geometry creation –

A 150 mm by 150 mm by 150 mm cubic concrete specimen was modelled. Using a scripting technique, coarse aggregates with randomly dispersed sizes between 8 and 16 mm were created inside this cube. To depict the diverse internal structure of concrete, aggregates were included in the mortar matrix. Discrete coarse aggregate inclusions and a continuous mortar phase made up the final geometry. For additional analysis, both geometries were merged and imported into ANSYS Workbench.

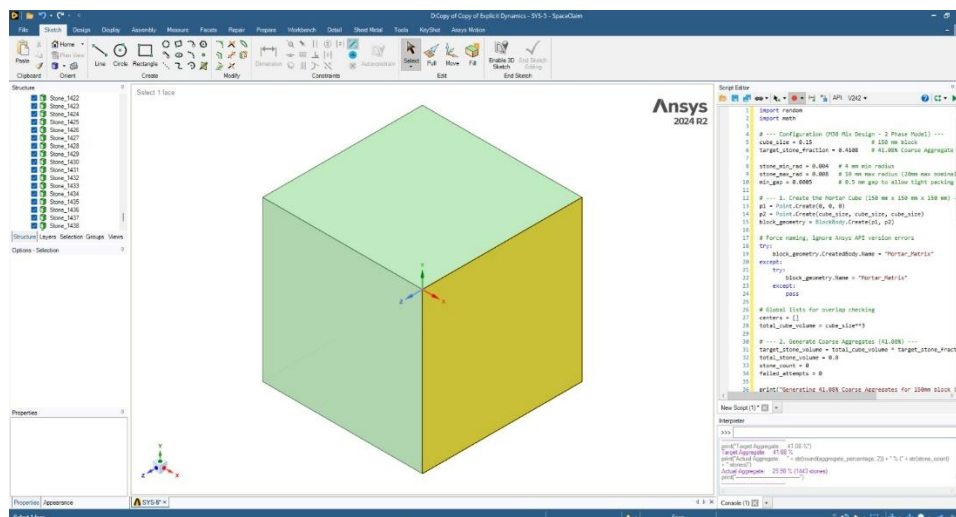


Figure 1 Generating the geometry of a 150 mm cube

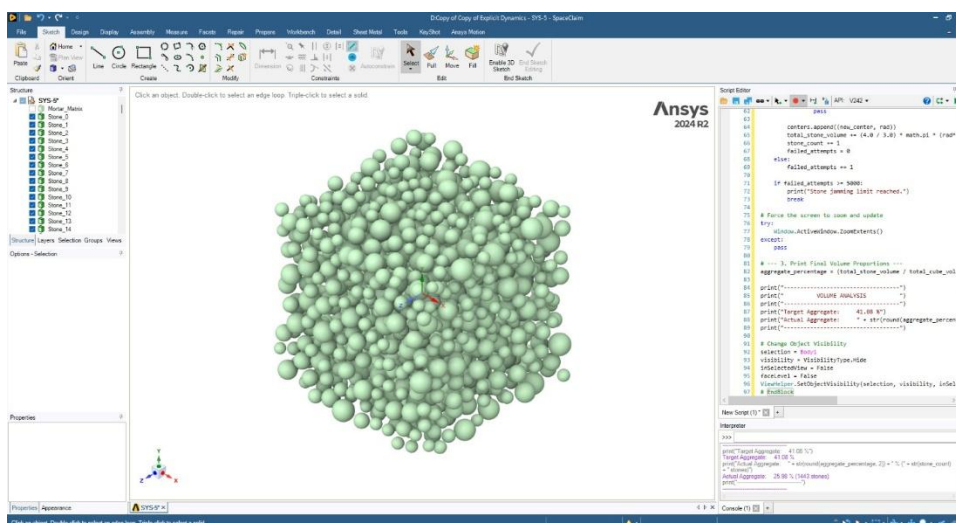


Figure 2 Generating coarse aggregate of 8-12 mm diameter

Material Properties -

Table 3 Material Properties

	Young's Modulus (E)	Poisson's ratio (v)	Density (ρ)
Mortar	28060 MPa	0.20	2100 kg/m ³
Coarse Aggregate	60000 MPa	0.17	2700 kg/m ³

	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Density	2700	kg m ⁻³		
4	Isotropic Elasticity				
5	Derive from	Young's Modulus and Poisson...			
6	Young's Modulus	60000	MPa		
7	Poisson's Ratio	0.17			
8	Bulk Modulus	3.0303E+10	Pa		
9	Shear Modulus	2.5641E+10	Pa		

Figure 3 Properties of Aggregate given to ANSYS

	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Density	2100	kg m ⁻³		
4	Isotropic Elasticity				
5	Derive from	Young's Modulus and Poisson...			
6	Young's Modulus	28060	MPa		
7	Poisson's Ratio	0.2			
8	Bulk Modulus	1.5589E+10	Pa		
9	Shear Modulus	1.1692E+10	Pa		

Figure 4 Properties of Mortar given to ANSYS

Effective density of concrete block = $(0.74)(2100) + (0.26)(2700) = 2256 \text{ kg/m}^3$

Meshing –

The combined geometry was discretised using a finite element mesh with an average element size of 2 mm. A sufficiently fine mesh was used to accurately capture wave propagation and interaction with aggregates.

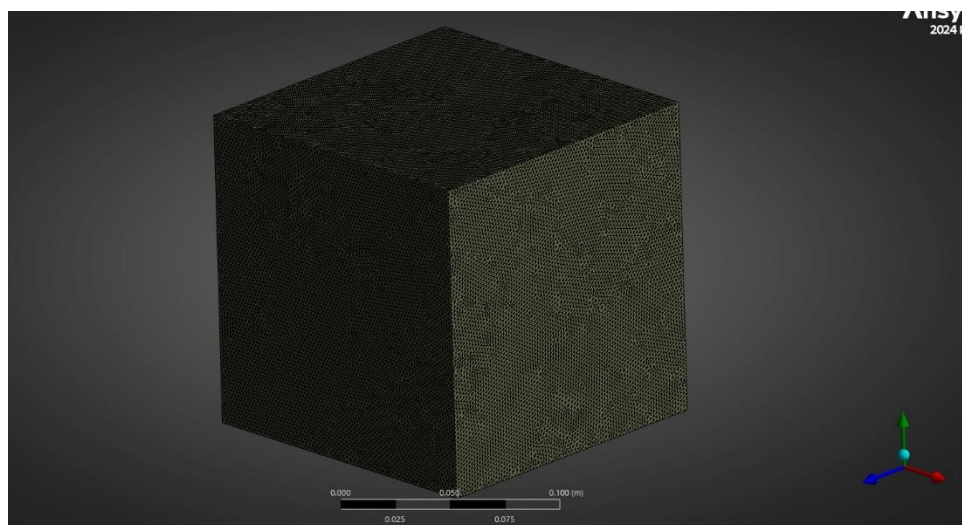


Figure 5 Meshing of a cube

Results Observed –

The velocity-time response recorded at the probe location was used to determine the arrival time of the first wave. This arrival time represents the effective travel time of the ultrasonic wave through the concrete specimen. And we got an arrival time of 42.5 μs ; after that, we can calculate the apparent UPV of the cube by dividing the path length by the observed time.

$$\text{UPV} = 150\text{mm}/42.5\mu\text{s}$$

$$\text{UPV} = 3529.41 \text{ m/s}$$

The simulated UPV closely matches our experimental data, indicating that our ANSYS model is well aligned with our practical blocks.

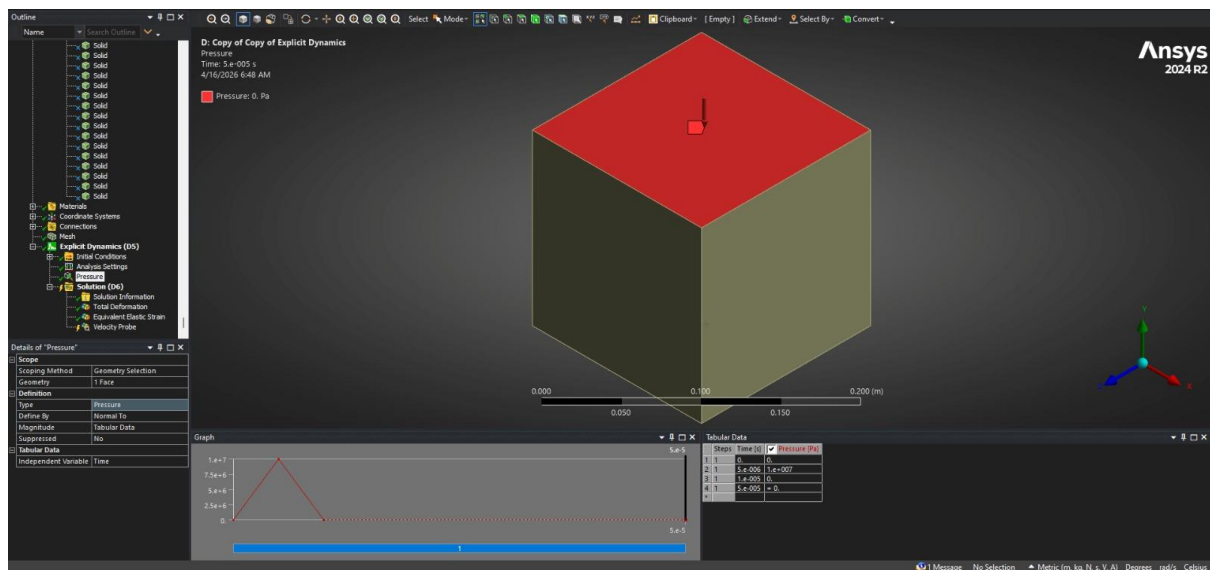


Figure 6 Wave Propagation from one face of the cube

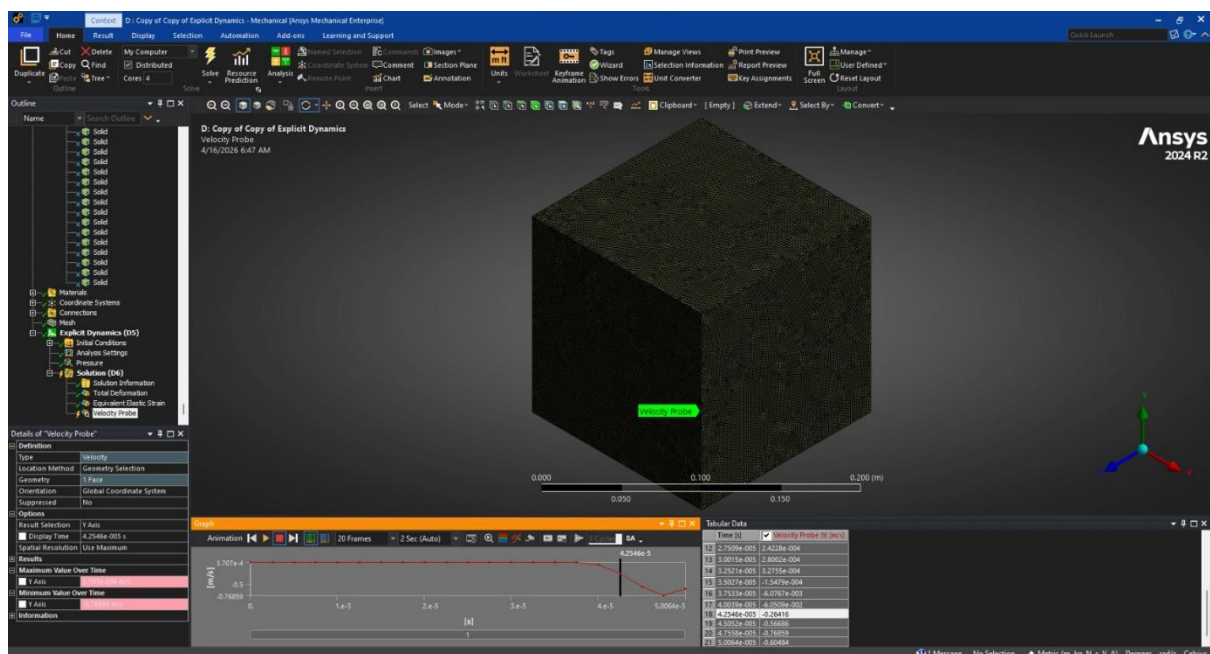


Figure 7 Observed results of the simulation

Observations made for the Young's modulus of Concrete –

The effective Young's modulus of concrete can be estimated using micromechanical models when the elastic properties of mortar and aggregates are known. Common models include the Voigt (upper bound) and Reuss (lower bound) approaches, as well as the more refined Hashin–Shtrikman (HS) bounds, which provide tighter, more realistic estimates for heterogeneous materials. In the present ANSYS model, concrete is represented as a mortar matrix with randomly distributed spherical aggregates of higher stiffness ($E_{agg} > E_{mortar}$). Since the stiff aggregates are dispersed within a continuous softer phase without forming a connected network, the effective elastic behaviour of the system is best approximated by the Hashin–Shtrikman lower bound, which corresponds to such inclusion-based microstructures.

Calculations –

Hashin-Shtrikman requires bulk modulus (K) and Shear Modulus (G) before calculating the Young's modulus.

$$K = \frac{E}{3(1-2\nu)} \quad G = \frac{E}{2(1+\nu)}$$

Then H-S bound on K –

$$K^{HS-} = K_m + \frac{V_f}{\frac{1}{K_a - K_m} + \frac{3V_m}{3K_m + 4G_m}}$$

Then H-S bound on G –

$$G^{HS-} = G_m + \frac{V_f}{\frac{1}{G_a - G_m} + \frac{6V_m(K_m + 2G_m)}{5G_m(3K_m + 4G_m)}}$$

Then the effective Young's modulus would be –

$$E_{eff} = \frac{9K^{HS-}G^{HS-}}{(3K^{HS-} + G^{HS-})}$$

Where:

K_a = Bulk Modulus of aggregate

K_m = Bulk Modulus of mortar

G_a = Shear Modulus of aggregate

G_m = Shear Modulus of mortar

V_f = Volume fraction of aggregate

V_m = Volume fraction of mortar

For our values from Table 3:

$$K_a = 30,303.03 \text{ MPa}$$

$$G_a = 25,641.03 \text{ MPa}$$

$$K_m = 15,588.89 \text{ MPa}$$

$$G_m = 11,691.67 \text{ MPa}$$

$$V_f = 0.26$$

$$V_m = 0.74$$

Now we can put these values into the formula of K^{HS-} and G^{HS-} and their values are coming out to be –

$$K^{HS-} = 18,424.2 \text{ MPa}$$

$$G^{HS-} = 14,208.4 \text{ MPa}$$

And now the effective Young's modulus would be –

$$E_{\text{eff}} = 33.9 \text{ GPa}$$

Formula for effective Poisson's ratio -

$$\nu_{\text{eff}} = \frac{3K^{HS-} - 2G^{HS-}}{2(3K^{HS-} + G^{HS-})}$$

From here, Poisson's ratio is 0.193.

Formula for Young's modulus using UPV –

$$E = \frac{\rho V^2 (1 + \nu)(1 - 2\nu)}{1 - \nu}$$

In this formula, we know the values which are –

$$V = 3529.41 \text{ m/s}$$

$$\rho = 2256 \text{ kg/m}^3$$

$$\nu = 0.193$$

$$E = 25.52 \text{ GPa}$$

Conclusion –

The paper combined both numerical simulation and experimental testing to examine the propagation of ultrasonic waves in heterogeneous concrete. Good-quality concrete was indicated by Ultrasonic Pulse Velocity (UPV) tests on laboratory-cast concrete cubes, which yielded values of 3416–3768 m/s. By explicitly simulating randomly dispersed coarse particles within a mortar matrix, a meso-scale numerical model was created in ANSYS. The simulated UPV value of roughly 3529 m/s was consistent with the experimental results. This demonstrates that the fundamental physics of wave propagation in heterogeneous media may be captured by the numerical model.

This study's main finding is that UPV does not directly correspond to compressive strength or stiffness because internal heterogeneities, such as aggregate distribution, impedance contrast, and scattering effects, significantly affect wave propagation. The comparison of the effective Young's modulus derived using various methods made this even clearer. The UPV-based dynamic modulus was substantially lower at 25.52 GPa, but the Hashin–Shtrikman (HS) technique produced a value of roughly 33.9 GPa.

These numbers differ because the underlying assumptions of the two approaches are fundamentally different. The Hashin-Shtrikman model is a static homogenization technique that provides an upper bound on stiffness by assuming perfect bonding between phases and an idealised material distribution. The UPV-based modulus, on the other hand, is determined from dynamic wave propagation, in which energy loss processes, scattering, wave-path distortion, and microstructural discontinuities all influence the measured velocity. Because of this, the UPV approach captures the material's effective dynamic response, which is usually lower than the idealised elastic modulus predicted by homogenization models.

A limitation of the present study is that the numerical model could not incorporate a maximum aggregate volume fraction of approximately 26% due to constraints on geometry generation in ANSYS. This is lower than the typical aggregate content in real concrete and may influence the simulated wave-propagation behaviour and the resulting UPV values.

Overall, this experiment shows that concrete's diverse internal structure significantly influences the propagation of ultrasonic waves. UPV is a helpful non-destructive method for evaluating the quality of concrete; it should not be used as a direct indicator of material characteristics without taking wave physics and heterogeneity into account. A more accurate non-destructive assessment of concrete structures is enabled by combining experimental results with numerical simulation, which provides a deeper understanding of the constraints and an appropriate interpretation of UPV.

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